

Anthropometric Determinants Influencing Paddler Performance: A Review

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ABSTRACT

Sports have been playing a dynamic role in maintaining fitness and improving physical skills & abilities since long. Sports are utilized by people for both recreation and competitiveness. Competitiveness is one of the prerequisites of sports and moreover, competitive sports not only involve contests, but advance as a central tenet as well that an athlete or a team will continually seek advancement to a higher level. Body shape and body type can determine how well a player can perform in a particular sport and most desirable parameters for a particular sport can be determined to ensure highest results in sports with the help of anthropometry. Anthropometry is the science of obtaining scientific measurements of human body. Anthropometric measurements have been used since long to associate physiological, cultural, psychological and physical properties and correlates of human body. The present paper throws light on the anthropometric determinants that may influence performance of a player in paddle sports (canoeing & kayaking). The overview of previous studies and essentiality of future research in these sports are discussed.

Keywords: Sports, anthropometry, kayaking, canoeing, anthropometric determinants.

INTRODUCTION

Human body has been of great interest to scientists since long. Man has always tried to understand what he is surrounded with and this has enabled him to divide and characterize objects into different groups. Characterization of objects can make one understand the characterization and division of human body with the help of different sciences. Jimenez et al. (2016) state that physical

and functional anthropology, macroscopic anatomy and basic & applied psychology intertwine creating sometimes other sciences including *kinanthropometry* for understanding body structures and how they are organized and affect people. Kinanthropometry refers to the dynamic relationship & quantitative interface between human structure and function.

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Ross et al. (1972) said kinanthropometry is a scientific discipline that studies the body size, proportionality, performance of movement, body composition and principal functions of the human body. Kinanthropometry is closely related to physical science and forms the base to contribute as a scientific interest to basic and applied research. It is also closely related to sports science, sports medicine, human biology, physical anthropology and ergonomics as well. The discipline of anthropometry and falls under the umbrella of kinanthropometry and is a matter of great interest to sports scientists (Jimenez, et al., 2016). It is within the above-mentioned sciences where the underlying review is discussed.

Anthropometry and Sports

Anthropometry plays an important role in the industrial design, clothing design, ergonomics and architecture where statistical data about the distribution of body dimensions are used. When the role of anthropometric measurements in sports is concerned, it was Tanner who opened an exciting new field for the benefit of athletes. He applied anthropometry to sports by measuring, photographing and X-raying about 350 athletes during the Commonwealth games and Rome Olympic Games in 1960. He had shown that such studies would help in a scientific selection of athletes for different events in improving their performances more effectively on scientific lines and also to a

great extent seeing the limits to which they might succeed. Since then, anthropometry is used as an applied discipline in sports for assessing the athlete performance and its betterment.

Kayaking and Canoeing

Boats have acted as a mode of transport since time immemorial. Evidences from early human migrations suggest that boats have been serving humans since prehistoric times. In earlier times, boats were used for travelling, hunting, fishing, fighting, trade, commerce, etc. and various models and types of boats were built and used depending upon the living conditions and ecological setup of humans. Despite fulfilling human needs since times, boats are used today for sporting and recreational purposes. Boats have experienced many developmental changes and progress with time.

Kayaking and canoeing are based on skill, endurance, strength and power that enable a paddler to stay on the boat and pull it towards the finishing point. Kayak strokes follow the same basic concept as canoe strokes: when the paddle is planted in the water and the paddler pulls, she is pulling self and canoe to the paddle. The power in kayak strokes comes from the push of the upper hand, twist of the torso, and pull of the lower hand. The use of all three will provide smooth, quick and strong strokes.

There are some differences between sprint kayaking and canoeing. In kayak,

the paddler is seated and the force is transmitted to the boat through the seat and the foot-rest, whereas canoeists are in a high kneeling position (upon one knee) in the boat. Furthermore, kayakers propel by means of a double-blade paddle alternately on both sides of the boat and steering is accomplished with a rudder attached with cords to a tiller controlled by the feet, while canoeists do it with a single-blade paddle always on the same side of the canoe and use steering strokes to control their direction (Alacid et al., 2015). Despite many differences between kayaking and canoeing, there is a similarity between the two which lies in its stroke. Boat lean helps kayakers feel the stability of their craft. It occurs when a paddler pulls one knee up, while simultaneously pushing down with the opposite hip and keeping his body weight above the kayak. This transfers weight to the hip and allows the paddler to balance on that hip. This will move the craft underneath him. When going forward, kayakers punch out with their upper arms and pull back with their lower arms. This creates a need for kayakers to rotate their body, while keeping their trunk and shoulders facing their hands.

Anthropometric determinants in Kayaking and Canoeing

Anthropometry has been playing an important role in determination of the possible prerequisites for competitive performance in sports and paddle sports especially canoeing & kayaking are not

untouched by this discipline. On the basis of the body build profile of athletes, the most desirable body parameters for a particular sport can be determined that would ensure attainment of the highest sports results. In the above context, researches between anthropometry and performance has increased over the past years (Gabbett and Georgieff, 2007; Lopez Plaza et al., 2017), still ample amount of research is unavailable. Body composition and anthropometry play an important role for athletic performance in paddling.

When paddle sports are concerned, both kayaking and canoeing involve upper body muscles to paddle the boat and both the sports require explosive power and endurance throughout the event. According to Lundstorm et al., (2019), paddling requires high values for maximal aerobic and anaerobic capacities in combination with a high upper-body and trunk musculature as well and strength which is regardless of being male or female. On the basis of review of pertinent literature of the concerned field, the author is of opinion that following anthropometric determinants may influence performance of a player in paddle sports:

Height: The measurement of height is a standard component of most of the fitness assessments. Height is an important attribute for many sports and there are advantages of both below and above average height. Data from London

Olympics (2012) and Rio Olympics (2016) shows that a taller height of the paddlers is observed among the Olympians who participated in London Olympics (2012) and Rio Olympics (2016). Moreover, numerous researchers have found that tall height has been an advantage to many sports including paddle sports. In a recent study among slalom canoeists, Busta et al. (2018) say that elite C1 competitors should be of average body height (175-180cm) which is based on their findings among elite and sub-elite athletes along with their comparison with other studies.

Body Weight: Measuring body weight in sports is important for assessing body composition and for monitoring changes in weight following diet and exercise. Excess body weight can raise adverse condition for an athlete and may slow down his performance in most of the sports (except weight category). Hence, optimum weight of an athlete according to a sport is an important determinant to perform well. In paddle sports, all canoe and kayak boats are required to meet certain criteria in terms of their size, shape, weight and length. Increased weight will cause the boat to sit deeper in water, increasing wetted area, total frictional resistance, and wave drag, thereby increasing the resistance that must be overcome by the paddler to propel the boat forward. Hence, weight of a paddler has to be at optimum level that may go in coordination with the size,

shape, weight and length of the boat (Lundstorm et al., 2019). In their study, Busta et al. (2018) say that the weight of the canoe slalom players should be 75-80kg which is based on their findings among elite and sub-elite athletes along with their comparison with other studies.

Sitting height: Sitting height gives a measure of the length of the trunk. The participant in a kayak is in a seated position with their legs inside the cockpit and extended in an anterior position (Michael et al., 2009). Kayaking involves a straight sitting to maintain proper balance while paddling until a player finishes up with race. This allows a broader range of movement and increase in strength to perform strokes and maneuvers. It is to say here that sitting height plays an important role in kayaking. According to Broomfield and Lauder (2015), raising a seat could have a potential positive impact on efficiency if utilised for female kayakers. However, the question of how big the seat raise should be remains. They are of opinion that having too big a seat raise for taller participants may unbalance them due to the centre of gravity being too high. The seat raise for smaller paddlers may need to be larger in order to produce better control over the boat. It is not incorrect to say here that increased sitting height in a player or raising the seat height of the kayak would affect the player performance, although it will be different for all individuals depending upon their

anthropometric make up and different degree of seat raise.

Body composition and body fat percentage: Body fat (adipose tissue) is a significant component and an important prerequisite of human body. But, excess body fat percentage in a player hinders the performance as it does not contribute to muscular force production. Studies have shown that excess of body fat adversely affects the sports performance of a player. Low body fat is an indispensable component of sports performance and minimum body fat percentage can excel the player performance and help a player do well (Busta et al., 2018). In their study, Messias et al. (2018) mention that fat percentage has possibly an important role in performance of elite kayakers. In addition, their study has also found a significant correlation of body fat and performance. Hence, it would not be incorrect to mention here that there should be a lower percentage of body fat in a player for a better performance.

Muscle Mass: The upper body build in kayakers is commonly recognized as one of the most important parameters affecting their sports results. Kayakers have a significant body muscle mass and small mean body fat mass. According to Tesch (1983), Fry & Morton (1991), Ackland et al., (2003), Rynkiewicz & Rynkiewicz (2010) and Busta et al (2018), trunk muscle mass is one of the significant factors affecting sports

results. Van Someren & Palmer (2000) illustrate in their study that a well-developed upper body musculature and the expression of muscular power differentiate between international and national-level kayakers. The international-level kayakers in their study were characterised by significantly greater upper body measurements and a significantly higher degree of mesomorphy than the national-level subjects. Hence, ample amount of muscle mass and well developed body musculature is required among paddlers for better performance.

Somatotype: Somatotyping is a system of classifying body types in terms of three categories: *endomorph*, *mesomorph* and *ectomorph*. Endomorphs have lot of body fat and muscle and they gain weight easily. Mesomorphs are athletic, strong and solid whereas ectomorphs are lean and long with less body fat and a little amount of muscle. Previous studies have shown predominant presence of mesomorphic element among slalom kayakers (Vaccaro and Gray, 1984; Ridge et al, 2007; Busta et al, 2018; Messias et al, 2018). Furthermore, study by Lopez Plaza et al (2017) confirm the presence of superior size of kayakers, more robust body mass and compact morphology as that of canoeists. Hence, the researcher is of opinion that mesomorphic component with small amount of ectomorphic element would be the best category of

somatotype for kayakers and canoeists for better performance as this will constitute little amount of muscle and athletic, strong and solid body build.

Knee height: It is a measurement of the distance from the base of the foot to the top of the knee. Knee height is correlated with stature, and is sometimes used to estimate height in bedridden patients. The proportion of knee height compared to leg length or sitting height may be interesting in a sporting context. Specific body proportions are important attributes for many sports. Kneeling increases the canoe's stability by lowering the center of gravity. It allows paddlers to use thigh and trunk muscles more efficiently in conjunction with their arms and shoulders. The researcher is of opinion that knee height, thigh length and leg length do matters for paddle performance and they may affect the performance.

Hand span: Measuring the hand span is a simple test for determining size of the hand. The hand size and span is advantageous for the sports that involve catching, gripping or tackling. Since, kayaking and canoeing involves gripping and holding the paddles firmly, a good hand length, hand breadth and hand size would influence a player's performance.

An overview from previous studies

Determinants, other than those discussed above like body build, length of the thigh, forearm & upper arm, biacromial breadth (shoulder width), arm span, chest,

skinfolds, and hips as well play an important role in influencing sports performance among paddlers. Cressie et al. (1986) in their study reveal that canoeists are characterised by very strong skeletal build, tallness, large body mass, long upper limbs, muscularity of the chest and upper limbs and athletic build.

In recent decades, data from anthropometric measurements on male and female elite canoe and kayak paddlers, report less body fat and increased upper-body and arm musculature that is related to performance and more successful paddling (Lundstorm et al., 2019). Studies on paddlers exhibit that there are distinct anthropometrical attributes in higher-level kayak athletes (Fry and Morton, 1991; Aitken and Jenkins, 1998; Ackland et al., 2003). Focusing the anthropometrical attributes, Ackland et al. (2003) reported in their study that elite sprint kayakers had well-developed upper body musculature, increased stature, narrow hips (for males), low skinfold values, increased sitting height and bi-acromial diameter in comparison to their sub elite counterparts. It is hence clear from their approach that biacromial breadth and increased sitting height influences the paddle performance which could be an innovative step for future researches among paddlers. Moreover, some researchers have shown that higher-level sprint kayak athletes typically have increased thigh, forearm and upper arm

length and lower skinfolds compared to both active university students and general population (Aitken and Jenkins 1998, Ackland et al., 2003). In addition to the above works, notable research by Messias et al. (2018) suggest that the anthropometric requirement in canoe slalom is specific characterized by low body fat percentage and large circumferences of upper limbs.

Within a particular sport, there exist various disciplines or playing positions with specific demands that require different approaches in training and are associated with different physical and morphological characteristics. Lundstorm et al. (2019) state that there is literature describing anthropometric features of international sprint canoe and kayak athletes competing in the 500 and 1000m events. Significant differences exist in height, body mass, sitting height and upper-extremity and forearm circumferences between international-versus national-level competitors in the 500 and 1000m events. In addition, significant differences have also been reported among junior national, state, and novice volleyball players for their stature, standing reach stature, skinfold thickness, spike jump, vertical jump, agility, and estimated maximal aerobic power (Gabbett and Georgieff, 2007; Mielgo-Ayuso et al., 2015). In their study, Alacid et al. (2015) state that one of the significant findings emerging from their study is that young kayakers were

heavier, taller, with greater sitting height, arm span, arm length, upper body breadths and girths than canoeists. These differences may be explained by the continued need for physical development in kayakers, in order to remain competitive, whereas young canoeists need to place a much greater emphasis on the development of their technical ability. Elite sprint kayak athletes tend to be taller and heavier than their sub elite level counterparts (Fry and Morton, 1991; Aitken and Jenkins, 1998; Ackland et al., 2003). These studies reveal that relatively similar anthropometrical determinants are a requisite for determining sports performance in both kayaking and canoeing. It is to say that the above studies can be carried out in future for finding out the similarities and dissimilarities between kayakers and canoeists.

CONCLUSION

The present study is an overview of the previous researches among the kayakers and canoeists. The literature review indicates clearly that there is a paucity of the anthropometric data in the investigation of kayaker and canoeist profiles. Studies among kayakers and canoeists and studies between them as well are to be carried out at massive level to compare their determinants and find out similarities & differences between players of these sports. A comprehensive anthropometric description and profiling of athletes shall help coaches to

understand their player and furthermore, would be helpful in talent identification. Furthermore, interdisciplinary research including significant focus of anthropometry is the need of the hour for exploring more anthropometric determinants for competitive performance in kayaking and canoeing having intervention from nutritionist, strength & conditioning expert, exercise physiologist, bio-mechanist, physiotherapist, psychologist and other

sports scientists. This shall rather be more helpful for intensive observation of athletes, extensive case studies and rigorous interpretations of the results.

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